

THERMODYNAMICS

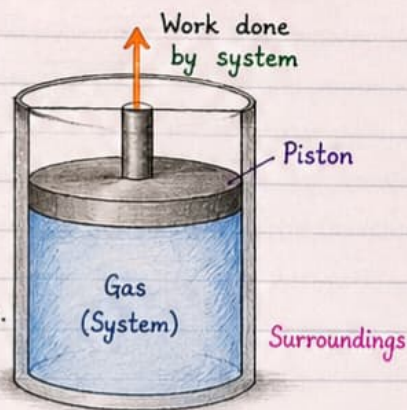
Key Idea

Thermodynamics explains how heat and work are related to each other.

Thermodynamics is the branch of physics that deals with the relationship between **heat**, **work** and other forms of **energy** and their interconversion.

BASIC CONCEPTS

- **System** → The part of the universe which is under study.
- **Surroundings** → Everything outside the system.
- **Boundary** → The surface separating the system from the surroundings.
- **Universe** → System + Surroundings.



IMPORTANT TERMS

- **Heat (Q)** → Energy transferred between system and surroundings due to temperature difference.
- **Work (W)** → Energy transferred when a force causes displacement of the boundary of the system.
- **Internal Energy (U)** → Total microscopic energy of the system. It includes kinetic + potential energies of molecules.
- **State** → Condition of a system described by measurable properties like P , V , T .

NOTE

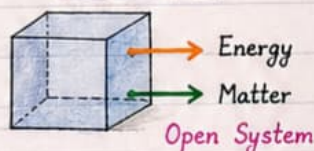
Heat and work are **not** properties of the system. They depend on the **path** taken between two states.

THERMODYNAMIC SYSTEM

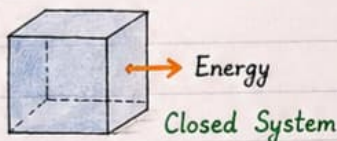
On the basis of exchange of **energy** and **matter**, systems are of three types:

- 1 **Open System** → Can exchange both **energy** and **matter** with surroundings.
Example → Boiling water in open vessel.
- 2 **Closed System** → Can exchange **energy** but **not** matter.
Example → Gas in a sealed cylinder with movable piston.
- 3 **Isolated System** → Can exchange neither **energy** nor **matter** with surroundings.
Example → Thermos flask.

3D VIEW



Open System



Closed System



Isolated System



When a system changes state, energy can be transferred as **heat** or **work**.



HEAT AND WORK

1. HEAT (Q)

- Heat is energy transferred between system and surroundings due to **temperature difference**.
- Heat flows from **higher** temperature body to **lower** temperature body.
- SI unit → **Joule (J)**

Example:

When hot water is poured in a cup, it loses heat and the cup gains heat.



2. WORK (W)

- Work is done when a **force** causes **displacement** of the boundary of the system.
- Work is said to be done **BY** the system when the system expands.
- Work is done **ON** the system when it is compressed.
- SI unit → **Joule (J)**

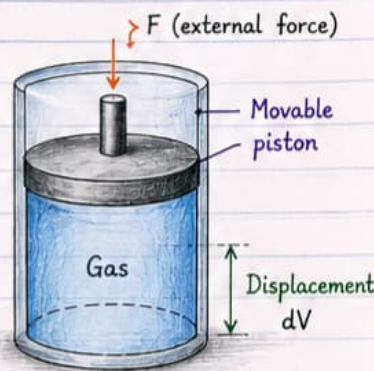
Formula (Boundary Work)

For a small displacement dV

$$dW = P_{\text{ext}} dV$$

For finite change from V_1 to V_2

$$W = \int_{V_1}^{V_2} P_{\text{ext}} dV$$



FIRST LAW OF THERMODYNAMICS

It is the law of **conservation of energy** applied to thermodynamic processes.

Statement → The change in internal energy of a system is equal to the heat supplied to the system **minus** the work done by the system.

$$\Delta U = Q - W$$

ΔU → Change in internal energy

Q → Heat supplied to the system

W → Work done by the system

Example:

When gas expands in a cylinder, if 50 J of heat is supplied and the gas does 20 J of work, then

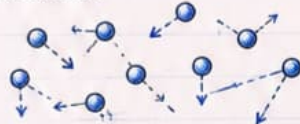
$$\Delta U = 50 - 20 = 30 \text{ J}$$

TYPES OF THERMODYNAMIC PROCESSES

	Process	Condition	Work (W)	Example	P-V Graph
①	Isothermal	$T = \text{constant}$	$W = nRT \ln \frac{V_2}{V_1}$	Slow expansion of ideal gas in contact with heat bath.	
②	Adiabatic	$Q = 0$	$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$	Rapid compression or expansion with no heat exchange.	
③	Isochoric	$V = \text{constant}$	$W = 0$	Heating gas in a rigid container.	
④	Isobaric	$P = \text{constant}$	$W = P(V_2 - V_1)$	Expansion of gas under constant pressure.	

MICROSCOPIC VIEW

Temperature is related to the average **kinetic energy** of molecules.



Increase in temperature → molecules move faster.

Remember!

- ΔU is a state function.
- It depends only on initial and final states, not on the path taken.

SECOND LAW OF THERMODYNAMICS

It gives the **direction** of natural processes.
No process is possible whose sole result is the transfer of heat from a colder body to a hotter body without any external work.

Key Idea

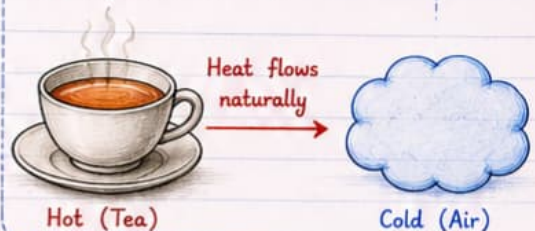
Nature always prefers disorder.
Useful energy gradually degrades into less useful forms and spreads out.

Statements (Equivalent)

- (i) It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a **colder** body to a **hotter** body.
- (ii) It is impossible to construct a **perpetual motion** machine of the second kind.
- (iii) Heat cannot, of itself, pass from a **colder** body to a **hotter** body.

Example :

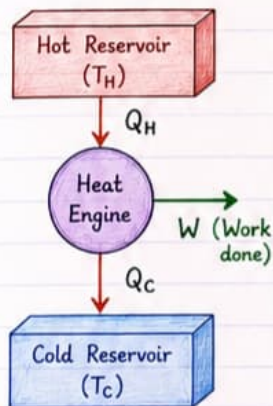
Heat flows from hot tea to air, not from air to tea.



HEAT ENGINES AND REFRIGERATORS

1. HEAT ENGINE

A **heat engine** is a device that converts heat into work continuously.



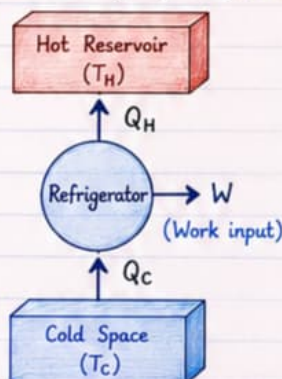
Efficiency (η)

$$\eta = \frac{\text{Work output}}{\text{Heat input}} = \frac{W}{Q_H}$$

For real engine, $\eta < 1$
(Some heat must be rejected to cold reservoir.)

2. REFRIGERATOR

A refrigerator removes heat from a cold space and rejects it to the surroundings by doing **external work**.



Coefficient of Performance (β)

$$\beta = \frac{\text{Heat removed from cold space}}{\text{Work input}} = \frac{Q_C}{W}$$

For ideal (Carnot) refrigerator,
 $\beta_{\max} = \frac{T_C}{T_H - T_C}$

Important Note

★ A real heat engine and a real refrigerator can never be 100% efficient.

Remember!

Heat engine → Heat to Work
Refrigerator → Work to move Heat

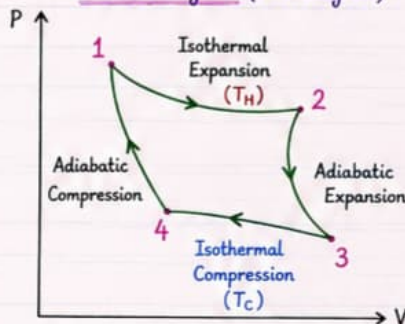
CARNOT ENGINE (IDEAL HEAT ENGINE)

- Works between two reservoirs at temperatures T_H (hot) and T_C (cold).
- Working substance undergoes a reversible cycle.
- Maximum possible efficiency among all engines working between T_H and T_C .
- Efficiency of Carnot engine :

$$\eta_{\max} = 1 - \frac{T_C}{T_H}$$

(T in Kelvin)

Carnot Cycle (P-V Diagram)



Steps of Carnot Cycle

1 → 2	Isothermal Expansion at T_H	Heat absorbed (Q_H) by the system
2 → 3	Adiabatic Expansion ($Q = 0$)	Temperature falls from T_H to T_C
3 → 4	Isothermal Compression at T_C	Heat rejected (Q_C) to cold reservoir
4 → 1	Adiabatic Compression ($Q = 0$)	Temperature rises from T_C to T_H

Example

A Carnot engine operates between 127°C and 27°C . Find its maximum efficiency.

Solution :

$$T_H = 127 + 273 = 400 \text{ K}$$

$$T_C = 27 + 273 = 300 \text{ K}$$

$$\eta_{\max} = 1 - \frac{T_C}{T_H} = 1 - \frac{300}{400} = 1 - 0.75 = 0.25 = 25\%$$

Quick Fact

The Carnot engine is an ideal engine. It cannot be realised in practice, but it sets the upper limit of efficiency. ★★

CHAPTER 11 : THERMODYNAMICS

ENTROPY (S)

- Entropy is a **state function** that measures the degree of **disorder** or randomness of a system.

- For a reversible process,

$$dS = \frac{\delta Q_{rev}}{T}$$

- SI unit $\rightarrow J K^{-1}$ (Joule per Kelvin)

Example :

Heat Q is supplied reversibly to an ideal gas at constant T .

Entropy change,

$$\Delta S = \frac{Q_{rev}}{T}$$



Key Idea

Entropy of an isolated system never decreases. It increases for natural processes and stays constant for reversible processes.

THIRD LAW OF THERMODYNAMICS

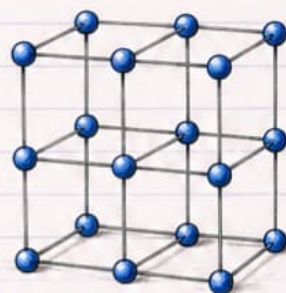
- It states that the entropy of a perfectly crystalline substance approaches **zero** as the temperature approaches **absolute zero**.

$$S \rightarrow 0 \quad \text{as} \quad T \rightarrow 0 K$$

- It is impossible to reach absolute zero temperature by any finite number of operations.

Example :

The entropy of a perfect crystal of a substance is zero at 0 K.

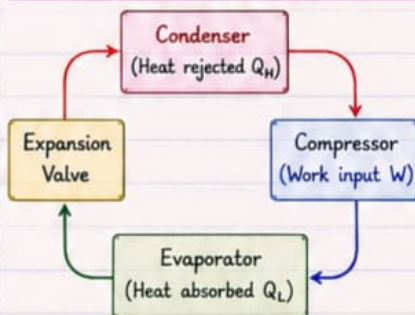


Perfect Crystal at 0 K

APPLICATIONS OF THERMODYNAMICS

- Steam Engine** - Converts heat energy of steam into mechanical work.
- Refrigerator** - Transfers heat from a colder region to a hotter region using external work.
- Air Conditioner** - Removes heat from indoor space and releases it outside.
- Power Plants** - Heat from fuel is used to produce steam which drives turbines to generate electricity.

Refrigeration Cycle (Simple)



Remember!

Thermodynamics helps us find the **direction** of natural processes and the maximum possible **useful work**.

SUMMARY OF FORMULAS

- First law : $\Delta U = Q - W$

- Work (small change) : $dW = P_{ext} dV$

- Work (finite change) : $W = \int_{V_1}^{V_2} P_{ext} dV$

- Heat (Reversible) : $dQ_{rev} = T dS$

- Entropy (Reversible) : $dS = \frac{\delta Q_{rev}}{T}$

- Carnot efficiency :

$$\eta_{max} = 1 - \frac{T_C}{T_H}$$

- Coefficient of Performance (Refrigerator) :

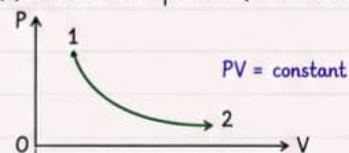
$$\beta = \frac{Q_C}{W} = \frac{T_C}{T_H - T_C} \text{ (ideal)}$$

- Entropy at absolute zero :

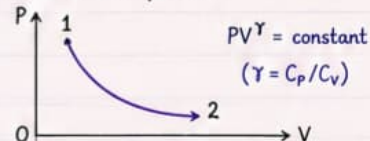
$$S \rightarrow 0 \text{ as } T \rightarrow 0 K$$

Typical P-V Diagrams

- Isothermal Expansion ($T = \text{constant}$)



- Adiabatic Expansion



NCERT Important Points

- ✓ Heat and work are path functions.
- ✓ Internal energy and entropy are state functions.
- ✓ Heat cannot be completely converted into work in a cyclic process.
- ✓ Entropy of an isolated system never decreases.

ONE LINE REVISION

- Energy is conserved.
- Heat flows from hot to cold.
- Work is done by a force.
- Entropy measures **disorder**.
- Nature prefers disorder.

Think Beyond!

Thermodynamics does not tell us how fast a process occurs, but only whether it can occur and in which direction.

- End of Chapter 11 -
Keep Practicing! -